

THESIS DEFENSE SPEECH AND BOARD Q&A PREPARATION

Hybrid Adaptive Transformer Differential Protection: An Intelligent DWT-LSTM Framework

Prepared for	Final Thesis Defense - B.Sc. in EEE
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Delivery target	20 minutes, two presenters, slide-by-slide

Rehearsal Notes

- Estimated spoken script length: 2,396 words. At 125-135 words per minute, this fits the 20-minute target with small pauses.
- Use the slide as the visual anchor. Do not read every bullet; explain the reason behind each slide.
- When a result is marked draft or indicative, say it as a validation trend and avoid claiming field certification.
- Presenter 1 covers the problem, objectives, model, implementation, and features. Presenter 2 covers dataset coverage, LSTM, deployment, results, comparison, conclusion, and Q&A transition.

20-Minute Run of Show

Part	Slides	Target	Purpose
Opening and motivation	1-6	4:10	Establish why 87T needs intelligent supervision.
Framework and implementation	7-16	5:25	Explain DWT, LSTM motivation, Simulink model, logging, and dataset structure.
Learning and relay decision	17-24	5:20	Show dataset coverage, LSTM training, ONNX deployment, and trip behavior.
Comparison and closing	25-33	5:05	Defend novelty, results, objectives, COs, conclusion, and future work.

Slide-by-Slide Defense Speech

Delivery guidance: The timing labels are rehearsal targets. If a slide has dense text, explain only the central idea and move on.

Slide 1: Title

Presenter 1 | Target time 0:00-0:35

Good morning respected chairman, honorable members of the thesis board, our supervisor, teachers, and guests. We are Md Moinul Haque and Md Reyaz Uddin from the Department of EECE, MIST. Today we are presenting our final thesis titled Hybrid Adaptive Transformer Differential Protection: An Intelligent DWT-LSTM Framework. In simple words, our work asks one protection question: how can a relay trip fast for a real transformer internal fault, but remain secure during inrush and external-fault CT saturation?

That balance between fast tripping and secure blocking is the theme of the whole presentation.

Slide 2: Outline

Presenter 1 | Target time 0:35-1:00

I will follow the outline shown here. First, I will explain the protection problem and why conventional transformer differential relays can mis-operate. Then I will summarize the literature gap, objectives, proposed framework, and implementation. After that, my teammate will discuss the dataset, classifier, Simulink deployment, results, comparison, course outcome mapping, conclusion, and future work.

Slide 3: Overview of Differential Protection using 87T Relay

Presenter 1 | Target time 1:00-1:45

Transformer differential protection is based on Kirchhoff's current law. Under normal operation, current entering the protected zone should equal current leaving it after ratio and vector-group compensation. The relay computes a differential current and compares it with a restraint current. The dual-slope

characteristic makes the relay sensitive at low through-current but more secure during heavy external faults. Ideally, high differential current means an internal fault. The difficulty is that some non-fault events can also create apparent differential current.

Slide 4: Research Motivation

Presenter 1 | Target time 1:45-2:45

This matters because power transformers are among the most expensive and critical substation assets. A correct internal-fault trip must be issued very quickly, normally within one cycle, which is less than 20 milliseconds at 50 hertz. A false trip can interrupt supply and cause cascading problems, while a missed trip can damage the transformer and create fire risk. Conventional 87T struggles mainly with magnetizing inrush and CT saturation. Inrush can reach several times rated current and look like a fault. Modern cores may have low second harmonic content, so harmonic blocking becomes unreliable. CT saturation during external faults can also distort secondary current and create false differential current. So the real challenge is not only speed. The relay must be fast, dependable, and secure at the same time.

Slide 5: Comparative and Associative Literature

Presenter 1 | Target time 2:45-3:25

The literature evolved from fixed harmonic restraint, to adaptive pickup and dual-slope bias, then to wavelet features, shallow machine learning, and more recently deep learning. The gap we identified is not simply accuracy. Prior methods often miss one practical requirement: temporal behavior, wide operating conditions, real-time deployment, or a deterministic fallback. Our work combines DWT time-frequency features with LSTM temporal modeling, while keeping the conventional 87T element as the protection anchor.

Slide 6: Research Objectives

Presenter 1 | Target time 3:25-4:10

The primary objective is to design and validate a hybrid adaptive differential protection scheme. We built a high-fidelity Simulink model of a 300 MVA, 230 over 11 kV, Yd1 transformer with CT saturation, inrush, and fault injection. Then we developed an adaptive 87T element with compensation, trained a DWT-LSTM classifier for normal, inrush, external, and internal events, and integrated both into a hybrid relay. Finally, we evaluated dependability, security, operating time, and robustness, while comparing against selected baselines.

Slide 7: Proposed Method / Model Framework

Presenter 1 | Target time 4:10-5:00

The full framework has six stages. First, six CT currents from the HV and LV sides are acquired. Second, Yd1 vector-group and CT-ratio compensation produce the differential and restraint quantities. Third, a db4 wavelet decomposition extracts multi-resolution information from a 20 millisecond window. Fourth, statistical features are formed, including energy, entropy, standard deviation, maximum value, mean absolute deviation, kurtosis, skewness, plus differential and restraint currents. Fifth, the LSTM classifies

the event. Finally, the decision logic trips only when the 87T pickup and internal-fault probability agree. This means the classical relay still detects the electrical abnormality, while the AI checks whether that abnormality is truly an internal fault.

Slide 8: Why LSTM Over Simpler Classifiers?

Presenter 1 | Target time 5:00-5:40

A single-window classifier, such as an ANN or SVM, sees only one snapshot. Inrush and internal faults can look similar at the beginning because both may produce high current and strong transients. The difference is in time evolution. Inrush usually starts large and decays over several cycles. An internal fault remains severe and persistent. LSTM is suitable because it keeps memory of previous windows and learns this evolution rather than judging one isolated waveform.

Slide 9: Implementation - Simulink Plant Model

Presenter 1 | Target time 5:40-6:15

For implementation, we modeled a 230 kV to 11 kV, 300 MVA, Yd1 transformer in Simulink. The source, transformer, CTs, and fault blocks are arranged to generate both internal and external fault conditions. The CT ratios are selected so that the secondary quantities are compatible after compensation. We also swept different fault types and fault resistances, so the relay is tested under more than one ideal condition.

Slide 10: Implementation - Test Console and Logging

Presenter 1 | Target time 6:15-6:45

To make the evaluation reproducible, we used a ControlPanel test harness. It automates normal condition, inrush, several internal faults, and selected external faults. For every run, the trip signal, differential current, restraint current, log files, and MAT files are stored. This avoids manual cherry-picking and allows batch dataset generation for training and result analysis.

Slide 11: Normal Condition and Magnetizing Inrush

Presenter 1 | Target time 6:45-7:05

This slide shows representative normal and inrush cases from the test console. The key point is that normal operation should never trip, while inrush must be blocked even if the differential current is temporarily high. This is exactly where a pure threshold-based relay can become insecure.

Slide 12: External Fault Cases

Presenter 1 | Target time 7:05-7:25

Here we show external three-phase and A-B faults. The fault is outside the protected transformer zone, so the correct decision is no trip from the transformer differential element. However, CT saturation during high external-fault current may produce waveform distortion, so this scenario is essential for security testing.

Slide 13: Internal Fault Cases

Presenter 1 | Target time 7:25-7:50

This slide shows internal fault cases such as A-G, A-B, and three-phase faults. These are the events where dependability matters most. The relay must trip quickly because the fault is inside the protected zone and continued current can create severe thermal and mechanical stress on the transformer winding.

Slide 14: Realism Injection

Presenter 1 | Target time 7:50-8:20

We did not train only on clean ideal waveforms. We added measurement noise, CT gain mismatch, ADC quantization error, and random fault inception angles. These non-idealities are important because protection devices work with imperfect measurements. If the model is trained only on ideal data, it may perform well offline but fail under real relay-like measurement conditions.

Slide 15: DWT-Based Feature Extraction

Presenter 1 | Target time 8:20-9:10

Now we come to the feature extraction step. We selected Daubechies-4 wavelet because it is compact, transient-sensitive, and widely used in power-system disturbance analysis. Wavelet decomposition separates the current waveform into frequency bands, so high-frequency fault transients and lower-frequency inrush components can be compared. From each level we compute energy and statistical descriptors. This converts raw current samples into a feature vector that is physically meaningful and easier for the LSTM to learn from.

In other words, DWT acts like an intelligent signal microscope. It does not replace protection logic; it gives the learning model cleaner evidence about where the transient energy appears.

Slide 16: Dataset Generation

Presenter 1 | Target time 9:10-9:35

The dataset includes the four operating classes: normal, inrush, internal fault, and external fault, which are then mapped into trip and no-trip decisions. The purpose is balanced protection learning: the model must recognize faults, but it must also learn when not to trip. Before final submission, the displayed class totals and trip-label totals should be checked for consistency, but the intended classification structure is clear.

Handoff: Presenter 1: This completes the model and dataset-generation setup. My teammate will now explain dataset coverage, the LSTM classifier, deployment, results, and conclusion.

Slide 17: Dataset Parameter Space Coverage

Presenter 2 | Target time 9:35-10:15

Thank you. I will continue from the dataset coverage. A protection model is useful only if it generalizes beyond one narrow case. Therefore, the dataset covers fault resistance, point-on-wave angle, sensor non-idealities, background noise, and different event timings. This prevents the LSTM from memorizing

a fixed waveform shape. Instead, it is forced to learn the underlying behavior of internal faults, external faults, inrush, and normal operation.

Slide 18: LSTM Classifier and Decision Logic

Presenter 2 | Target time 10:15-10:55

The LSTM input is a sequence of feature vectors extracted from consecutive windows. The network has two LSTM layers with dropout to reduce overfitting, followed by a softmax output over four classes. Training used Adam optimization, class-weighted cross-entropy, early stopping, cross-validation, and data augmentation. The important design idea is that the classifier is not used alone; its internal-fault probability is used as a supervised input to the relay decision.

Slide 19: AI Deployment in Simulink

Presenter 2 | Target time 10:55-11:30

After training in PyTorch, the model was exported to ONNX and converted for use in Simulink through the Deep Learning Toolbox Predict block. This is important because many AI protection papers stop at offline classification. Our workflow places the model back into the simulation environment, so the relay logic, plant model, and classifier interact in a closed-loop style rather than only in a spreadsheet of features.

Slide 20: Hybrid Relay Logic

Presenter 2 | Target time 11:30-12:25

The hybrid relay has three outcomes. First, if differential current is above a very high hardware setting, the relay trips immediately. This is the safety override for catastrophic internal faults. Second, for ordinary operation, the relay trips when the 87T element picks up, the LSTM assigns high internal-fault probability, and the decision remains consistent across consecutive windows. Third, if 87T picks up but the LSTM identifies inrush or external-fault saturation, the relay blocks. So the AI is a security supervisor, not an uncontrolled replacement for the classical relay.

This architecture is defensible because a single model error cannot directly command a trip; it must pass through the conventional protection condition.

Slide 21: Training Result - Main Observations

Presenter 2 | Target time 12:25-13:05

The training results show stable convergence, high accuracy, and strong separation between fault and non-fault cases. The confusion matrix indicates that most cases are correctly classified, and sensitivity remains high across different inception angle regions. The main limitation is high-impedance faults: as fault resistance becomes very large, the fault current becomes closer to load or non-fault behavior. This is why the hybrid relay retains the 87T structure and does not depend only on the classifier.

Slide 22: Training Result - Statistical Confidence

Presenter 2 | Target time 13:05-13:35

The bootstrap analysis gives confidence intervals for accuracy, precision, recall, F1 score, AUC-ROC, and average precision. These metrics show that performance is not based on one lucky split. The key message is balanced performance: high precision means few false trips, and high recall means few missed internal faults. In protection, both are important, but a missed internal fault is usually the more dangerous failure.

Slide 23: Trip Behaviour Comparison

Presenter 2 | Target time 13:35-14:15

This slide compares conventional 87T behavior with the hybrid approach. The conventional relay can trip incorrectly during inrush or CT-saturated external faults because both create apparent differential current. The hybrid relay blocks these cases when the LSTM recognizes their temporal and wavelet signatures, while still tripping for true internal faults. The extra AI decision time remains below the one-cycle protection requirement.

Slide 24: Robustness Validation

Presenter 2 | Target time 14:15-14:55

Robustness was tested under changes in fault resistance, measurement noise, CT gain mismatch, residual flux, and random inception angle. The slide reports that dependability and security remain at or above the target level across the sweep. Since some table values are marked as indicative target values pending the full sweep run, we should present them carefully: the trend is that the proposed hybrid scheme improves security and speed compared with conventional harmonic restraint and other baselines.

This is also where we acknowledge practical limits. Simulation can show strong evidence, but final confidence for field use requires hardware-in-the-loop and real disturbance data.

Slide 25: Comparative Literature - Methodology Overview

Presenter 2 | Target time 14:55-15:30

Compared with representative literature, our work combines several design choices in one framework. Earlier methods may use wavelets, neural networks, or adaptive settings, but many lack temporal modeling, CT saturation sweep, deployment pathway, or hardware fallback. The novelty is the combination: DWT features, sequence learning, confidence-gated decision, and deterministic 87T override in a reproducible Simulink-based workflow.

Slide 26: Quantitative Performance Benchmark

Presenter 2 | Target time 15:30-16:10

The benchmark table summarizes accuracy, dependability, security, false positives, and trip time. The main interpretation is that the proposed scheme reduces false trips while maintaining fast internal-fault operation. We should emphasize relative behavior rather than overstate absolute numbers: the hybrid design improves security over conventional harmonic restraint and gives a clearing time within the one-cycle budget. If asked, we can explain that dependability is internal-fault detection rate, and security is false-trip prevention rate.

That distinction is important because a relay with high accuracy can still be unacceptable if its few errors are dangerous protection errors.

Slide 27: Gap Analysis and Novelty

Presenter 2 | Target time 16:10-16:50

This slide states the novelty in terms of design dimensions. The proposed framework uses a richer DWT-based representation, temporal modeling through LSTM, confidence-gated AND logic, and a deterministic hardware override. From a protection-engineering viewpoint, the most important point is responsibility: AI improves discrimination, but the safety-critical trip path is still connected to conventional relay logic. That makes the approach more defensible than a pure black-box relay.

Slide 28: Completion of Research Aims

Presenter 2 | Target time 16:50-17:20

Here we map each research objective to a deliverable. The Simulink plant, adaptive 87T compensation, DWT-LSTM classifier, hybrid relay subsystem, validation logs, and comparative analysis have all been completed. This slide is important because it connects the thesis promises with actual outputs, not only with final metric values.

Slide 29: Course Outcome Achievement (1 of 2)

Presenter 2 | Target time 17:20-17:45

For course outcomes, CO1 is addressed by identifying the protection problem and research gap. CO2 is addressed by designing the Simulink and Python pipeline, including compensation, DWT features, LSTM classification, and hybrid relay logic. CO3 is addressed through technical investigation, scenario sweeps, robustness testing, and comparison with baseline methods.

Slide 30: Course Outcome Achievement (2 of 2)

Presenter 2 | Target time 17:45-18:10

CO4 is covered by modern engineering tools such as MATLAB, Simulink, PyTorch, ONNX, and the Deep Learning Toolbox. CO5 is addressed through societal impact: reliable transformer protection reduces outages, equipment damage, and safety risk. CO6 is supported by honest reporting of limitations, and CO7 by teamwork, supervisor review, thesis writing, and oral communication.

Slide 31: Conclusion

Presenter 2 | Target time 18:10-18:55

In conclusion, this thesis delivered a simulation-based hybrid relay framework for transformer differential protection. DWT features capture time-frequency behavior, LSTM captures temporal evolution, and the conventional 87T element remains as a fail-safe anchor. The framework improves discrimination between normal operation, inrush, external faults, and internal faults under realistic disturbances such as CT saturation, noise, and CT mismatch.

The central contribution is the coordinated handshake between proven relay logic and intelligent classification.

Slide 32: Future Prospects

Presenter 2 | Target time 18:55-19:35

The immediate future work is hardware-in-the-loop validation using RTDS or OPAL-RT, testing with real COMTRADE disturbance records, optimizing the model for relay hardware such as FPGA or DSP, and studying cyber-perturbed measurements. Longer-term, the same idea can be extended to different transformer ratings, vector groups, and even other protection zones such as busbars, generators, and transmission lines.

Slide 33: Thank You

Presenter 2 | Target time 19:35-20:00

That concludes our presentation. The key message is that we are not replacing protection engineering with AI; we are using AI as an intelligent supervisor to improve security while retaining the dependability of the classical relay. Thank you for your attention. We welcome the committee's questions.

Board Q&A Preparation

Answer style: Start with the direct answer, then give one technical reason, then connect it back to protection dependability and security.

Protection Fundamentals

Q1. What is transformer differential protection?

Answer: It compares current entering and leaving the transformer protected zone after correcting CT ratio and vector group. If the corrected difference is high, the relay interprets it as an internal fault and trips.

Q2. What is ANSI 87T?

Answer: ANSI 87T is the transformer differential protection function. The number 87 means differential protection, and T specifies transformer application.

Q3. Why do we use restraint current?

Answer: Restraint current prevents false operation during high through-fault current. As through current increases, CT error and saturation risk increase, so the relay raises its operating threshold using a slope characteristic.

Q4. What is dependability and what is security?

Answer: Dependability means the relay trips when there is a real internal fault. Security means the relay does not trip for non-fault or external-fault conditions. A good transformer relay needs both.

Inrush, CT Saturation, and Fault Discrimination

Q5. Why does magnetizing inrush cause false differential current?

Answer: During energization, high current may flow from the source side to magnetize the transformer core while the other side may carry little or no current. To the differential relay, this looks like unequal currents even though there is no internal fault.

Q6. Why can second-harmonic restraint fail?

Answer: Classical inrush blocking assumes inrush contains strong second harmonic. Modern transformer cores and different residual flux conditions can reduce second harmonic content, so the fixed harmonic threshold may not reliably distinguish inrush from a real fault.

Q7. How does CT saturation create false trips?

Answer: For an external fault, primary currents should balance through the protected zone. If one CT saturates, its secondary current becomes clipped and no longer represents the primary current correctly. The relay then sees a false differential current.

Q8. What is the main difference between inrush and internal fault in time behavior?

Answer: Inrush is usually high at energization and then decays as the core flux settles. An internal fault remains dangerous and persistent until isolated. This time evolution is why sequence modeling with LSTM is useful.

DWT Feature Extraction

Q9. Why use DWT instead of FFT?

Answer: FFT shows frequency content but loses time localization. Transformer faults and inrush are transient events, so we need both time and frequency information. DWT gives multi-resolution time-frequency features.

Q10. Why choose db4 wavelet?

Answer: Daubechies-4 is compact and sensitive to sharp transients. It is also common in power-system disturbance studies, so it gives a practical balance between feature quality and implementation simplicity.

Q11. What does wavelet energy mean physically?

Answer: Wavelet energy measures how much signal activity exists in a frequency band. Internal faults and switching events distribute energy differently across bands, so band energy helps separate their signatures.

Q12. Why include statistical features like entropy, kurtosis, and skewness?

Answer: Energy alone may not capture the full waveform shape. Entropy measures distribution spread, kurtosis detects sharp heavy-tailed transients, and skewness captures asymmetry. Together they make the feature vector more descriptive.

LSTM and Hybrid Logic

Q13. Why LSTM instead of ANN or SVM?

Answer: ANN and SVM usually classify one window independently. LSTM reads a sequence and remembers earlier windows, so it can learn whether a transient is decaying like inrush or persisting like an internal fault.

Q14. Is the AI directly tripping the breaker?

Answer: No. The AI is used as a supervisory veto layer. The relay trips only when the conventional 87T element operates and the LSTM confirms an internal fault with sufficient confidence, except for the high-current hardware override.

Q15. Why use a confidence threshold such as 0.85?

Answer: The threshold prevents weak or uncertain classifications from causing a trip. It is a practical way to trade off security and dependability, and it can be tuned using validation data.

Q16. Why require consecutive windows?

Answer: A single noisy window may be misleading. Consecutive-window consistency reduces random misclassification and makes the decision more stable without exceeding the one-cycle operating requirement.

Q17. What is the purpose of the hardware override?

Answer: If differential current is extremely high, the relay should not wait for AI confirmation. The override ensures catastrophic internal faults can be cleared immediately and the system remains fail-safe if the AI block is unavailable.

Dataset, Simulation, and Deployment

Q18. Why is simulation-only validation acceptable for this thesis?

Answer: For undergraduate thesis scope, high-fidelity simulation is a practical first validation step. The work models transformer, CT saturation, faults, noise, mismatch, and random inception angles. However, field and HIL testing are correctly listed as future work.

Q19. How did you avoid overfitting?

Answer: The workflow uses varied event parameters, noise and CT mismatch augmentation, dropout, class-weighted loss, early stopping, and cross-validation. The dataset also includes multiple operating classes rather than only one fault type.

Q20. Why deploy through ONNX into Simulink?

Answer: ONNX lets a trained PyTorch model be used inside Simulink. This allows the trained classifier to interact with the relay model and plant simulation, which is closer to a real deployment workflow than offline scoring.

Q21. What are the limits of the present dataset?

Answer: It is generated from one transformer rating and vector group, and the final proof is simulation-based. Generalization to other transformers, real disturbance records, and hardware-in-the-loop testing is future work.

Results, Metrics, and Defensibility

Q22. How should we interpret the reported high accuracy?

Answer: Accuracy shows overall classification success, but protection decisions need more specific metrics. Dependability tells whether internal faults are detected; security tells whether false trips are avoided. Those two metrics are more important than accuracy alone.

Q23. Why does high-impedance fault performance degrade?

Answer: High fault resistance reduces fault current magnitude, so the waveform may look less severe and closer to load or non-fault conditions. This is a known protection challenge, and the hybrid relay reduces risk by keeping the 87T element in the loop.

Q24. Why claim real-time feasibility?

Answer: The decision is based on short windows, compact feature extraction, and LSTM inference that fits inside the one-cycle target in the Simulink evaluation. Final field proof still needs HIL and embedded relay testing.

Q25. What is the strongest contribution of the thesis?

Answer: The strongest contribution is architectural: combining classical 87T dependability with learned LSTM-based security supervision. This is more practical than making a black-box classifier the only trip decision-maker.

Q26. What should we say if asked about differing metric values in the slides?

Answer: State that classifier-only metrics, hybrid relay metrics, and indicative benchmark values refer to different evaluation views. For final defense, use the value on the presented slide, and clearly distinguish validation-set results from draft benchmark targets.

Novelty, Standards, and Future Work

Q27. What is novel compared with prior wavelet-LSTM work?

Answer: The novelty is not merely using wavelet or LSTM. It is the complete hybrid protection framework: DWT-based features, temporal sequence modeling, confidence gating, consecutive-window check, Simulink deployment, and deterministic 87T fallback.

Q28. Is this IEC-certified?

Answer: No. The thesis does not certify a commercial relay. It keeps a certified-style conventional protection element as the dependability anchor and proposes a research framework that would need HIL, relay-hardware, and standard-compliance testing before field use.

Q29. How can this be extended to real substations?

Answer: The next steps are COMTRADE data testing, HIL validation using RTDS or OPAL-RT, model optimization for DSP or FPGA, IEC 61850 sampled-value and GOOSE integration, and field pilot testing with utility data.

Q30. Can the method apply to other protection systems?

Answer: Yes. The idea of combining classical protection with an AI supervisory layer can be tested for busbar, generator, and line protection, especially where CT saturation, evolving faults, and transient discrimination are difficult.

Verify Before Defense

Purpose: These checks are included so the presentation remains defensible if the board asks for exact implementation values.

- Confirm whether the final deck should report 5-level db4 DWT with 37 features at 10 kHz, or the thesis text's compact 3-level energy-feature setup at 1.6 kHz.
- Check the dataset summary on Slide 16 because the displayed class totals and trip/no-trip counts appear internally inconsistent.
- Decide the final official performance statement: classifier-only accuracy, hybrid relay dependability/security, and benchmark table values should be clearly separated.
- If Slide 24 values remain labelled indicative or target draft, state them as trend/benchmark values rather than final certified values.
- Use careful wording for IEC compliance: the thesis proposes an IEC-aligned fail-safe architecture; it does not certify a commercial relay.

Final reminder: defend the architecture, not just the numbers. The strongest claim is that a classical 87T element provides the dependability anchor, while DWT-LSTM supervision improves security against inrush and CT-saturation false trips.